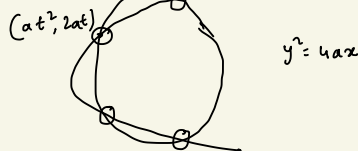


1) Parabola and a Circle:

$$y^2 = 4ax$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$



Maximum 4 points of contact.

$$(at^2)^2 + (2at)^2 + 2g(at^2) + 2f(2at) + c = 0$$

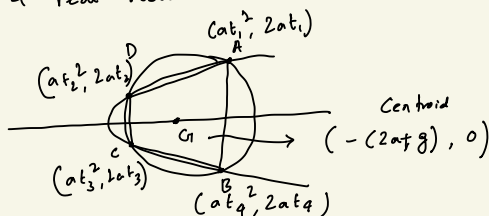
$$\Rightarrow t^4 + 4t^2 + \frac{2g}{a}t^2 + \frac{4f}{a}t + \frac{c}{a^2} = 0$$

$$\Rightarrow t^4 + (4 + \frac{2g}{a})t^2 + \frac{4f}{a}t + \frac{c}{a^2} = 0$$

Maximum 4 real roots

4 roots

t_1, t_2, t_3, t_4



$$\checkmark t_1 + t_2 + t_3 + t_4 = 0.$$

$$\checkmark t_1t_2 + t_1t_3 + t_1t_4 + t_2t_3 + t_2t_4 + t_3t_4 = 4 + \frac{2g}{a}$$

$$\checkmark t_1t_2t_3 + t_2t_3t_4 + t_3t_4t_1 + t_4t_1t_2 = -\frac{4f}{a}$$

$$t_1t_2t_3t_4 = \frac{c}{a^2}$$

$\rightarrow 0$

$$ABCD \text{ centroid : } \left(\frac{at_1^2 + at_2^2 + at_3^2 + at_4^2}{4}, \frac{2at_1 + 2at_2 + 2at_3 + 2at_4}{4} \right)$$

$$= \left(\frac{a}{4} (t_1^2 + t_2^2 + t_3^2 + t_4^2), \frac{2a}{4} (t_1 + t_2 + t_3 + t_4) \right)$$

$$= \left(\frac{a}{4} (t_1^2 + t_2^2 + t_3^2 + t_4^2), 0 \right)$$

$$[t_1^2 + t_2^2 + t_3^2 + t_4^2 = (t_1 + t_2 + t_3 + t_4)^2 - 2 \sum t_1t_2]$$

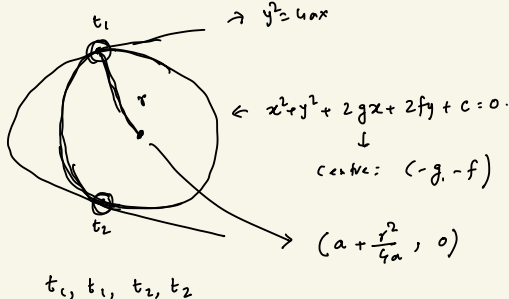
$$= 0 - 8 - \frac{4g}{a}$$

$$= - \left(8 + \frac{4g}{a} \right)$$

$$= \left(- (2a + g), 0 \right).$$

The centroid lies on the x-axis

□



$$2(t_1 + t_2) = 0 \Rightarrow t_1 + t_2 = 0$$

and,

$$(t_1 t_2)^2 = \frac{c}{a^2}$$

$$t_1^2 + t_2^2 + 4t_1 t_2 = 4 + \frac{2g}{a}$$

$$\Rightarrow (t_1 + t_2)^2 + 2t_1 t_2 = 4 + \frac{2g}{a}$$

$$\Rightarrow 0 + 2t_1 t_2 = 4 + \frac{2g}{a}$$

$$\Rightarrow (t_1 t_2)^2 = \left(2 + \frac{g}{a}\right)^2$$

$$\Rightarrow \frac{c}{a^2} = \left(2 + \frac{g}{a}\right)^2$$

and,

$$t_1^2 t_2 + t_1^2 t_2 + t_2^2 t_1 + t_2^2 t_1 = -\frac{4f}{a}$$

$$\Rightarrow 2t_1 t_2 (t_1 + t_2) = -\frac{4f}{a}$$

$$\Rightarrow \boxed{f=0}$$

$\therefore$ Centre on the x-axis

$$\text{radius} = r = \sqrt{g^2 - c}$$

$$4 + \frac{g^2}{a^2} + \frac{4g}{a} = \frac{c}{a^2}$$

$$\Rightarrow 4 + \frac{g^2 - c}{a^2} + \frac{4g}{a} = 0$$

$$\Rightarrow 4 + \frac{g^2}{a^2} + \frac{4g}{a} = 0$$

$$\Rightarrow 4a + \frac{g^2}{a} + 4g = 0$$

$$\Rightarrow \boxed{g = -\left(a + \frac{g^2}{4a}\right)}$$

$$t^4 + \left(4 + \frac{2g}{a}\right)t^2 + \frac{c}{a^2} = 0$$

$$\Rightarrow t^4 + 2\left(2 + \frac{g}{a}\right)t^2 + \left(2 + \frac{g}{a}\right)^2 = 0$$

$$\Rightarrow \left[t^2 + \left(2 + \frac{g}{a}\right)\right]^2 = 0$$

$$\Rightarrow t^2 = -\left(2 + \frac{g}{a}\right)$$

$$\Rightarrow t = \pm \sqrt{-\left(2 + \frac{g}{a}\right)}$$

$$\left\{ \begin{array}{l} t_1 = + \sqrt{-(2 + \frac{g}{a})} \\ t_2 = - \sqrt{-(2 + \frac{g}{a})} \end{array} \right\} \quad \checkmark$$

give the point of contacts.

□

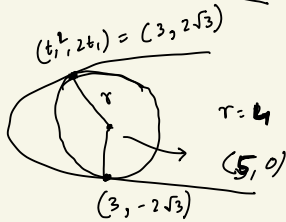
$$y^2 = 4x$$



Circular ring
radius = 4

$$y^2 = 4x$$

$$a = 1$$



$$a = 1$$

$$r = 4$$

$$f = 0$$

$$g = - \left(a + \frac{r^2}{4a} \right)$$

$$= - \left(1 + \frac{16}{4} \right)$$

$$= -5 \quad \Rightarrow \quad \boxed{-g = 5}$$

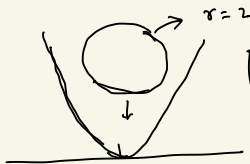
$$(-g, -f)$$

$$\begin{aligned} t_1 &= \sqrt{-(2 + \frac{g}{a})} = \sqrt{-(2 + \frac{-5}{1})} \\ &= \sqrt{-(2 - 5)} \\ &= \sqrt{3} \end{aligned}$$

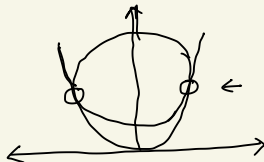
$$\therefore t_2 = -\sqrt{3}$$

□

H.W

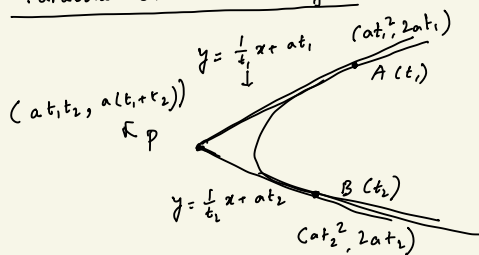


$$y = x^2$$



← coordinates?

Parabola and three tangents:

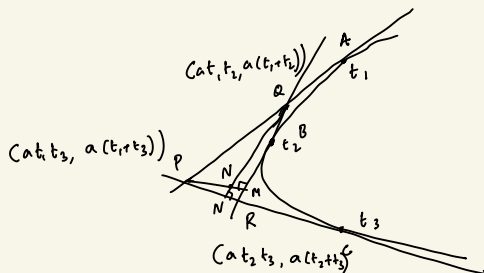


$$y^2 = 4ax$$

$$\begin{aligned} y &= \frac{1}{t_1}x + at_1 \\ (-) \quad y &= \frac{1}{t_2}x + at_2 \\ \hline \Rightarrow x \left(\frac{1}{t_1} - \frac{1}{t_2} \right) &= a(t_2 - t_1) \\ \Rightarrow x &= at_1t_2 \end{aligned}$$

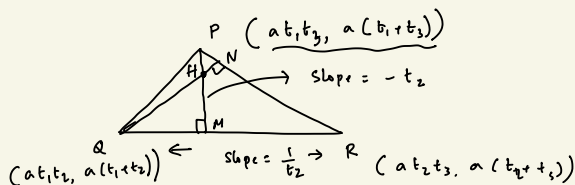
$$\begin{aligned} y &= \frac{1}{t_1} \cdot at_1t_2 + at_1 \\ &= a(t_1 + t_2) \end{aligned}$$

□



$$y^2 = 4ax$$

• orthocentre of $\triangle PQR$.



$$\text{Eqn of PM: } \frac{y - a(t_1 + t_3)}{x - at_1t_3} = -t_2$$

$$\Rightarrow y - a(t_1 + t_3) = -t_2x + at_1t_2t_3$$

$$\Rightarrow y = -t_2x + a(t_1 + t_3 + t_1t_2t_3)$$

$$\text{Eqn of QN: } y = -t_3x + a(t_1 + t_2 + t_1t_2t_3)$$

$$y = -t_2x + a(t_1 + t_3 + t_1t_2t_3)$$

$$y = -t_3x + a(t_1 + t_2 + t_1t_2t_3)$$

$$\Rightarrow 0 = (t_3 - t_2)x + a(t_1 + t_3 - t_1 - t_2)$$

$$\Rightarrow x = -a$$

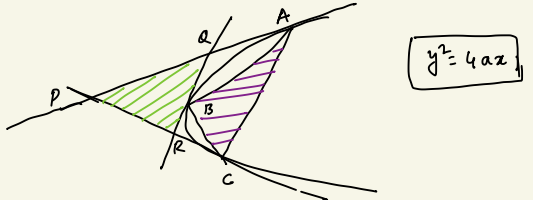
$$y = -t_2(-a) + a(t_1 + t_3 + t_1t_2t_3)$$

$$= a(t_1 + t_2 + t_3 + t_1 t_2 t_3)$$

or two centre : $(-a, a(t_1 + t_2 + t_3 + t_1 t_2 t_3))$

↑ lies on the directrix.

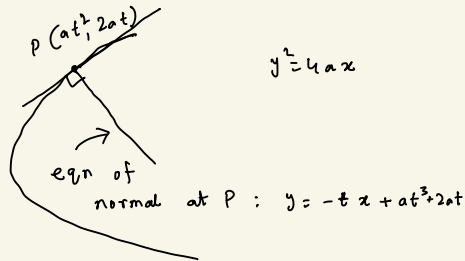
H.W.



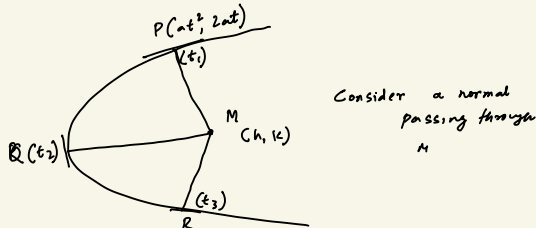
Qn The value of $\frac{[ABC]}{[PQR]}$ is

- A) 1 B) $\frac{1}{2}$ C) 2 D) $\frac{3}{2}$

Conormal Points :



0 Fix a point inside $y^2 = 4ax$



Consider a normal
passing through
M

eqn of PM : $y = -tx + at^3 + 2at$

$(h, k) \rightarrow \Rightarrow k = -th + at^3 + 2at$

$\Rightarrow at^3 + 2at - th - k = 0$

$\Rightarrow t^3 + 2t - \frac{h}{a}t - \frac{k}{a} = 0$

$\Rightarrow t^3 + \left(2 - \frac{h}{a}\right)t - \frac{k}{a} = 0$

roots:

t_1, t_2, t_3

Cubic in t .

max. 3 real solutions.

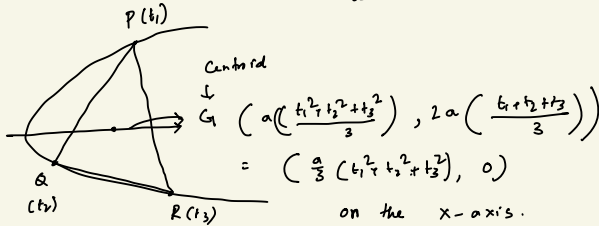
From any point inside the parabola, at max
3 tangents can be drawn on that parabola

P, Q, R are called conormal points.

$$t_1 + t_2 + t_3 = 0$$

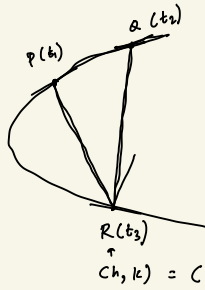
$$t_1 t_2 + t_2 t_3 + t_3 t_1 = 2 - \frac{4}{a}$$

$$t_1 t_2 t_3 = \frac{k}{a}$$



□

$$y^2 = 4ax$$



Normals at $P(t_1), Q(t_2)$
 meet again on
 the parabola at
 $R(t_3)$

Find the relation

btw

$$t_1, t_2, t_3$$

$P, Q, R \rightarrow$ Conormal points.

$$t_1 + t_2 + t_3 = 0$$

$$t_1 t_2 t_3 = \frac{k}{a} = \frac{2at_3}{a}$$

$$\Rightarrow t_1 t_2 = 2$$

Ans

Normal at t_1 :

$$y = -t_1 x + 2at_1 + at_1^3$$

$$\Rightarrow 2at_3 = -at_1 t_3^2 + 2at_1 + at_1^3$$

$$\Rightarrow 2t_3 = -t_1 t_3^2 + 2t_1 + t_1^3$$

$$\Rightarrow 2(t_3 - t_1) = t_1(t_1^2 - t_3^2)$$

$$\Rightarrow 2(t_3 - t_1) + t_1(t_3^2 - t_1^2) = 0$$

$$\Rightarrow (t_3 - t_1)(2 + t_1(t_3 + t_1)) = 0$$

$$\Rightarrow 2 + t_1 t_3 + t_1^2 = 0$$

Similarly,

$$2 + t_2 t_3 + t_2^2 = 0$$

$$(-) \quad t_1 t_3 + t_1^2 - t_2 t_3 - t_2^2 = 0$$

$$\Rightarrow t_3(t_1 - t_2) + (t_1 - t_2)(t_2 + t_3) = 0$$

$$\Rightarrow (t_1 + t_2 + t_3)(t_1 - t_2) = 0$$

$$\Rightarrow t_1 + t_2 + t_3 = 0$$

$$\Rightarrow t_3 + t_1 = -t_2$$

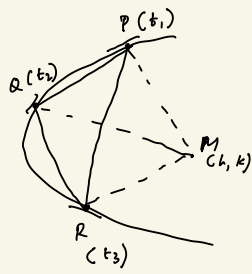
and,

$$2 + t_1(t_3 + t_1) = 0$$

$$\Rightarrow 2 + t_1(-t_2) = 0$$

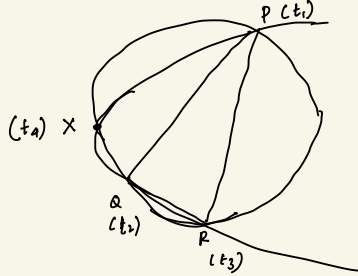
$$\Rightarrow 2 - t_1 t_2 = 0 \Rightarrow t_1 t_2 = 2$$

Prop



$$y^2 = 4ax$$

P, Q, R Conormal points.



Let, P, Q, R
be conormal
points.
Then the circumcircle
of $\triangle PQR$ passes
through the origin.

$$t_1 + t_2 + t_3 + t_4 = 0$$

as, P, Q, R are conormal: $t_1 + t_2 + t_3 = 0$

$$\Rightarrow t_4 = 0$$

4th point of contact: $(at_4^2, 2at_4) \equiv (0, 0)$.

H.W.

Let $P(x_1, y_1), Q(x_2, y_2), R(x_3, y_3)$ be
three conormal points. Find the value of

$$\frac{x_1 - x_2}{y_3} + \frac{x_2 - x_3}{y_1} + \frac{x_3 - x_1}{y_2}$$

_____ / _____